Bargaining over Cabinets: Multi-Agent RL for Portfolio Allocation in Coalition Governments

Arpan Jain (B23CS1089) Aradhya Mahajan (B23CS1088)

GitHub: `https://github.com/bot2048/cabinet-bargaining-marl`

1 Problem and findings

After an election without a majority, parties bargain over who governs and how ministries are split. Real coalition cabinets divide portfolios almost exactly in proportion to seats (*Gamson's law*): on 195 real cabinets, share $= 0.031 + 0.896\,s + 0.011\,F$ (s: seat share in the coalition, F: prime minister's party; R^2 0.93). The leading bargaining theory, Baron–Ferejohn (BF), predicts instead that the *formateur* (the party proposing the cabinet) captures a large premium. We model parties as reinforcement-learning agents in a multi-agent environment of government formation, rewarded only by their own office and policy payoffs (no fairness term), validate them against the exact BF equilibrium, and test which institutions remove the premium. Findings:

F1 Validated learner. In the BF game learned formateur shares match the closed form within 0.017 (10 seeds).

F2 Under proposal bargaining the premium never disappears; it changes currency. The base game gives a formateur coefficient $\hat{\gamma} = +0.249$. Private valuations or policy motives drive the *counted* premium to zero, but the formateur keeps it in the ministries it values most or in the policy-setting ministries.

F3 Real cabinets have no premium in any currency. Real prime-minister parties also hold the key ministries in proportion, and Gamson's law predicts real splits better than any learned agent or BF.

F4 The protocol is what matters. Under demand bargaining the agents' own cabinets follow Gamson's law with the small-party bias of real cabinets (slope 0.80, $\hat{\gamma} = -0.031$).

2 RL formulation

Environment (`cabinet/env.py`, `cabinet/config.py`): a partially observable stochastic game with n parties (seats w_i, majority quota) and m ministries. In a period without a cabinet a formateur $f \sim \rho$ proposes a winning coalition $C \ni f$ and an allocation $\mu : \{1..m\} \to C$; the cabinet forms if every partner accepts and pays per-period flows until the term ends or a member exits. Switchable mechanisms: repeated terms (M1), demand protocol (M2), private types (M4), heterogeneous values (M5), exits (M8), ministerial policy (M9), recognition rule (M10); level E0 is the exact BF game.

State $\mathcal{S}$. A *set* of n party tokens (26 features: seats, recognition probability, roles, offers, last Λ cabinets), m ministry tokens (8: salience, own valuation, policy dimension led), pair features, a 17-dim context and 8 decision features; no labels, so one permutation-equivariant network serves any n, m. **Action $\mathcal{A}$** (masked): the formateur includes/excludes each party (masks keep the coalition winning) and points each ministry to a member (10^{19} proposals in $n{-}1{+}m$ steps); voters accept/reject; members stay/exit; under M2 a party demands a share (1% grid) or closes. **Reward $\mathcal{R}$.** $r_i = \sum_{k:\mu(k)=i} \nu_{ik} - \lambda \frac{1}{d} \sum_h |y_h - z_{ih}|$ per governed period (value of own ministries minus policy loss; the holder of a lead ministry sets y_h), an exit bonus on leaving, 0 otherwise; discount δ. Rewards are individual.

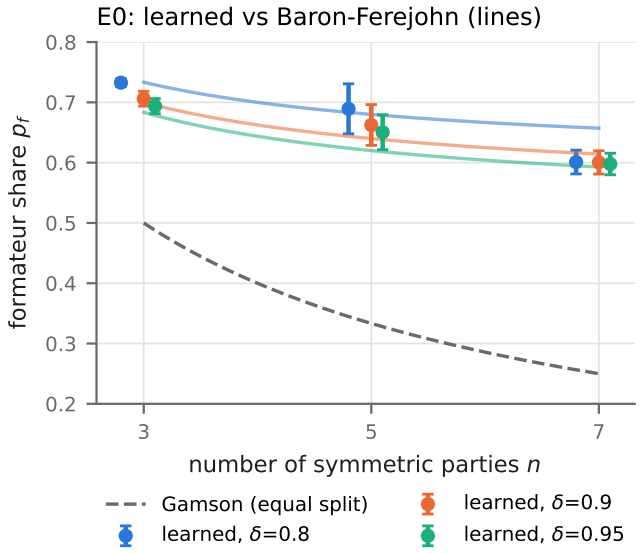


Figure 1: F1: learned formateur share (dots) vs the BF closed form (lines) in the symmetric game; dashed: Gamson's equal split.

Algorithm (`cabinet/policy.py`, `cabinet/ppo.py`): one shared attention network with masked autoregressive heads, trained by PPO with GAE on each agent's own decision sequence (semi-MDP); also MAPPO (central critic), fictitious self-play, PSRO (`cabinet/psro.py`) and a GRU memory. Baselines: the exact BF strategy from our solver (`cabinet/bf_solver.py`), Gamson, myopic and random agents (`cabinet/baselines.py`). Metrics (`cabinet/metrics.py`): regression (1) $p = \alpha + \beta s + \gamma F$, its analogues for value and policy-ministry shares, and the floor-adjusted Gamson distance $\tilde{D}_G$. Scale: $n \le 5$ (3–8 for real parliaments), $m = 6$, 500–2000 PPO iterations per run, ≥ 3 seeds (10 for the main cells).

3 Results

F1, validation. Learned shares follow the BF closed form (Fig. 1; mean gap +0.001, 6/9 cells within 0.02), including its decrease with patience δ; errors grow with n, where each voter's threshold is learned less precisely. The same code then produces all other results, so differences between configurations come from the institutions, not the learner.

F2, the premium changes currency. The base game keeps the BF premium ($\hat{\gamma} = +0.249$), and it grows with training (– after 4× more updates). Repetition, even with patient agents and GRU memory, exits and the recognition rule leave it in place (Fig. 2), as do MAPPO, fictitious self-play and PSRO (Table 1); a companion implementation of the same question by our team finds 0.20–0.24. Two mechanisms remove the *counted* premium: private types ($\hat{\gamma} = -0.036$) and ministerial policy (-0.027), and removing private types from the full game restores it. But the premium moves: with private valuations the formateur's share of what *it* values exceeds its seat share by +0.157 (base +0.147), and with policy it holds the policy-setting

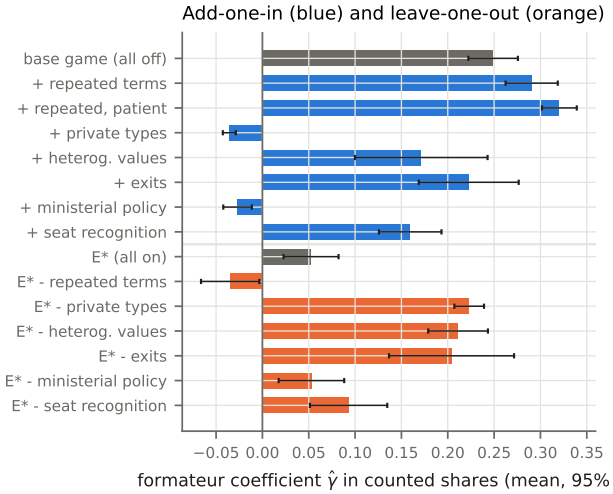


Figure 2: F2: formateur coefficient in counted shares when a mechanism is added to the base game (blue) or removed from the full game E* (orange); mean and 95% CI over seeds.

Table 1: The full game E* under different learners, memory and ministry counts: counted coefficient $\hat{\gamma}$, slope $\hat{\beta}$ and the formateur's premium in its own valuation Φ_ν.

	$\hat{\gamma}$	$\hat{\beta}$	Φ_ν	$\tilde{D}_G$
IPPO, window Λ=4	+0.052	+0.427	+0.143	+0.093
MAPPO (central critic)	+0.037	+0.460	+0.140	+0.085
Fictitious self-play	+0.063	+0.398	+0.150	+0.097
IPPO + GRU memory	+0.040	+0.420	+0.139	+0.097
IPPO, memoryless	-0.009	+0.444	+0.110	+0.096
IPPO, 12 ministries	-0.061	+0.417	+0.141	+0.121

ministries (coefficient +0.845 on its share of them). The formateur trades the number of ministries for their quality or for policy control.

F3, real cabinets. We imposed each real coalition and prime minister (Seki–Williams data linked to ParlGov) on policies trained only on random parliaments and compared their split with the real one (Fig. 3, 183 cabinets). Gamson's law errs by 0.052, BF by 0.144 (implied $\hat{\gamma} = 0.42$). Agents with private types err by 0.087–0.091 (implied $\hat{\gamma}$ 0.12–0.19), much closer to Gamson than the same agents with public types (0.131–0.167, $\hat{\gamma}$ 0.41–0.53, like BF), but every one is worse than Gamson's law (paired CI above +0.016). Is the real premium hidden in quality, as in our agents? No: on 131 cabinets the prime minister's party holds 0.589 of the key ministries (finance, foreign affairs, interior, defence, justice) against 0.573 of all ministers, a difference of +0.015 (95% CI -0.021 to +0.056). Real seat shares also fit better than voting power (R^2 0.93 vs 0.87), whereas our agents' shares follow voting power. When the agents choose the coalition themselves in 242 real parliaments, they pick the governing one in 21.7% of cases, like Riker's minimal-seat rule (19.7%), behind an ideological rule (31.7%).

F4, the protocol. Under demand bargaining (Morelli; each party in turn demands a share or closes a cabinet with earlier demanders) on the same random parliaments, the agents' own cabinets give share $= +0.095 + 0.80\,s + -0.031\,F$ (first mover +0.048; 2 seeds), close to the real $(0.031, 0.896, 0.011)$ including the small-party bias, while the matched proposal game gives $\hat{\gamma} = --$. In symmetric parliaments the learned demands still favour the first mover, a lock-in we are testing.

Reliability. A best response trained for 400 PPO iterations gains +0.015 per period against E*, but +0.037 and +0.088 against scripted Gamson and BF-heuristic profiles:

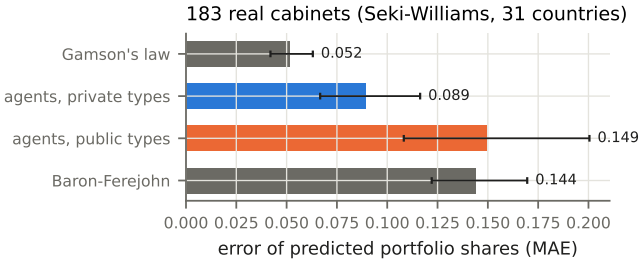


Figure 3: F3: predicting the portfolio split of real coalition cabinets, given the real coalition and prime minister; country-clustered 95% CIs.

the learned profiles are approximate equilibria at our budget, and the test has the power to tell. Cross-play between independent runs costs +0.0020 per period.

4 Implications and limitations

Self-interested learning under the closed-rule proposal protocol reproduces bargaining theory, not Gamson's law: whatever the information, memory, exits or learner, the formateur keeps a premium, at most moving it from the number of ministries to their quality or to policy. Real cabinets show no premium in counts or in key ministries, so their proportionality is not a premium in disguise. What changes the outcome is the protocol: when parties make demands instead of answering a formateur's proposal, learned allocations become seat-proportional. Limitations: scaled-down sizes; exploitability tests refute but do not prove equilibrium; real allocations count ministers (salience only through the key-ministry test); the prime minister's party stands in for the formateur. Coverage of the specification: `docs/SPEC_COVERAGE.md`.

Key code (beyond those above): `scripts/run_experiment.py`, `scripts/seki_eval.py`, `scripts/key_ministries.py`, `tests/test_core.py`.

References

[1] D. Baron, J. Ferejohn. Bargaining in legislatures. *APSR* 83(4), 1989.

[2] W. Gamson. A theory of coalition formation. *Am. Sociol. Rev.* 26(3), 1961.

[3] E. Browne, M. Franklin. Aspects of coalition payoffs in European parliamentary democracies. *APSR* 67(2), 1973.

[4] P. Warwick, J. Druckman. The portfolio allocation paradox. *EJPR* 45(4), 2006.

[5] S. Ansolabehere et al. Voting weights and formateur advantages in the formation of coalition governments. *AJPS* 49(3), 2005.

[6] J. Schulman et al. Proximal policy optimization algorithms. arXiv:1707.06347, 2017.

[7] C. Yu et al. The surprising effectiveness of PPO in cooperative multi-agent games. NeurIPS, 2022.

[8] J. Heinrich, D. Silver. Deep reinforcement learning from self-play in imperfect-information games. arXiv:1603.01121, 2016.

[9] H. Döring, P. Manow. ParlGov: Parliaments and governments database. parlgov.org.

[10] K. Seki, L. Williams. Updating the party government data set. *Electoral Studies* 34, 2014.

[11] M. Morelli. Demand competition and policy compromise in legislative bargaining. *APSR* 93(4), 1999.